

15-1c Natural Monopolies

An industry is a

natural monopoly (a type of monopoly that arises because a single firm can supply a good or service to an entire market at a lower cost than could two or more firms)

when a single firm can supply a good or service to the entire market at a lower cost than could two or more firms. A natural monopoly arises when there are economies of scale over the relevant range of output. [Figure 1](#) shows the average total costs of a firm with economies of scale. In this case, a single firm can produce any amount of output at the lowest cost. That is, for any given amount of output, a larger number of firms leads to less output per firm and higher average total cost.

Figure 1

Economies of Scale as a Cause of Monopoly

When a firm's average-total-cost curve continually declines, the firm has what is called a natural monopoly. In this case, when production is divided among more firms, each firm produces less, and average total cost rises. As a result, a single firm can produce any given amount at the lowest cost.

An example of a natural monopoly is the distribution of water. To provide water to residents of a town, a firm must build a network of pipes throughout the town. If two or more firms were to compete in the provision of this service, each firm would have to incur the fixed cost of building a network. Thus, the average total cost of providing water is lowest if a single firm serves the entire market.

We saw other examples of natural monopolies in [Chapter 11](#), when we noted that *club goods* are excludable but not rival in consumption. An example is a bridge used so

infrequently that it is never congested. The bridge is excludable because a toll collector can prevent someone from using it. The bridge is not rival in consumption because use of the bridge by one person does not diminish the ability of others to use it. Because there is a large fixed cost of building the bridge but a negligible marginal cost of additional users, the average total cost of a trip across the bridge (the total cost divided by the number of trips) falls as the number of trips rises. Hence, the bridge is a natural monopoly.

When a firm is a natural monopoly, it is less concerned about new entrants eroding its monopoly power. Normally, a firm has trouble maintaining a monopoly position without ownership of a key resource or protection from the government. The monopolist's profit attracts entrants into the market, and these entrants make the market more competitive. By contrast, entering a market in which another firm has a natural monopoly is unattractive. Would-be entrants know that they cannot achieve the same low costs that the monopolist enjoys because, after entry, each firm would have a smaller piece of the market.

In some cases, the size of the market is one determinant of whether an industry is a natural monopoly. Again, consider a bridge across a river. When the population is small, the bridge may be a natural monopoly. A single bridge can meet the entire demand for trips across the river at the lowest cost. Yet as the population grows and the bridge becomes congested, meeting the entire demand may require multiple bridges across the same river. Thus, as a market expands, a natural monopoly can evolve into a more competitive market.

QuickQuiz

1. Some government grants of monopoly power are desirable if they
 - a. curtail the adverse effects of cut-throat competition.
 - b. make industries more profitable.
 - c. provide incentives for invention and artistic creation.
 - d. save consumers from having to choose among alternative suppliers.
2. A firm is a natural monopoly if it exhibits ____ as its output increases.
 - a. increasing total revenue
 - b. increasing marginal cost
 - c. decreasing marginal revenue
 - d. decreasing average total cost

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